Business Requirements Document (BRD)

Product: Client Ticketing & SLA Management Software (working title) **Version:** 0.1 (Draft) | **Status:** For review **Prepared for:** Internal use first, then licensed to similar businesses

1. Purpose of this Document

This document defines the business needs, scope, and requirements for a ticketing (helpdesk) system. The system lets clients raise issues in a structured way and lets our team track, respond to, and resolve them within response times promised in each client's service package. It will first be used internally and then sold as licensed, installable software to similar businesses.

2. Background and Problem Statement

Clients currently report issues mainly by phone and informal messages. As a result:

- Calls are missed or forgotten and there is no record of the issue.
- There is no consistent procedure for who handles what, and by when.
- Management has no visibility into workload, delays, or recurring problems.
- Clients on higher-value packages do not get a visibly faster or more responsive experience.

We need a systematic, trackable process that gives clients a responsive experience in line with their package.

3. Business Objectives

ID	Objective	Measure of success
BO-1	Ensure every client issue is logged and acknowledged	100% of issues have a ticket; zero issues tracked only by phone
BO-2	Meet response and resolution commitments per package	At least 95% of tickets within SLA
BO-3	Give management real-time visibility	Dashboard in daily use by managers

ID	Objective	Measure of success
BO-4	Improve client satisfaction	Client satisfaction score at or above 4/5
BO-5	Build a reusable product for sale	80-90% of the codebase shared across customers; new customer installed in days, not months

4. Scope

4.1 In scope

- Client portal for raising and tracking tickets
- Ticket creation via portal, email, and (phase 2) WhatsApp
- Ticket lifecycle workflow, assignment, and escalation
- Package-based SLA engine with timers and breach alerts
- Internal team dashboard and reports
- Notifications to clients and staff
- Configuration layer for packages, SLAs, categories, statuses, branding, and templates
- Extension/module mechanism for customer-specific customization
- Installable, per-customer deployment with licensing

4.2 Out of scope (initial release)

- Multi-tenant SaaS hosting, self-service signup, and subscription billing
- Live chat and voice call integration
- AI-based auto-resolution or chatbots
- Native mobile apps (a responsive web app is in scope)

5. Stakeholders and User Roles

Role	Description	Key needs
Client	External customer of the business	Raise issues easily, see status, get timely updates

Role	Description	Key needs
Support Agent	Handles tickets	Clear queue, full history, SLA visibility
Team Lead / Manager	Oversees agents	Assignment control, escalations, performance reports
System Administrator	Configures the product	Manage users, packages, SLAs, branding, modules
Business Owner / Management	Strategic oversight	Dashboards, trends, SLA compliance
Software Customer (future)	Company buying the licence	Easy install, configurability, data ownership, support

6. Business Requirements (High Level)

- **BR-1:** Every client issue must be captured as a uniquely identified ticket with a full history.
- **BR-2:** Response and resolution targets must depend on the client's service package.
- **BR-3:** The business must be alerted before and after an SLA is breached, with escalation.
- **BR-4:** Clients must be able to see the status of their issues at any time and receive updates.
- **BR-5:** Management must be able to monitor workload, SLA compliance, and trends.
- **BR-6:** Business rules (packages, SLAs, categories, workflow) must be changeable without code changes.
- **BR-7:** Customer-specific needs must be addressed through plug-in modules without altering the core product.
- **BR-8:** The product must be installable on a customer's own infrastructure and protected by a licence.

7. Functional Requirements

7.1 Client Portal

ID	Requirement	Priority
FR-1	Client can log in securely and see only their own organisation's tickets	Must
FR-2	Client can raise a ticket with title, description, category, priority, and attachments	Must
FR-3	System issues a unique ticket number and confirmation immediately	Must
FR-4	Client can view ticket status, history, and replies, and add comments	Must
FR-5	Client can reopen a resolved ticket within a configurable window	Should
FR-6	Client can rate the resolution (satisfaction survey)	Should
FR-7	Client can see their package and its response-time commitments	Should
FR-8	Tickets can also be created by sending an email to a support address	Must
FR-9	Tickets can be created via WhatsApp	Could (phase 2)

7.2 Ticket Lifecycle and Assignment

ID	Requirement	Priority
FR-10	Default statuses: New, Assigned, In Progress, Waiting on Client, Resolved, Closed (configurable)	Must
FR-11	Tickets can be assigned manually or automatically by category, round-robin, or workload	Must
FR-12	Agents can add public replies and private internal notes	Must

ID	Requirement	Priority
FR-13	Every change is recorded in an audit trail (who, what, when)	Must
FR-14	Tickets can be merged, linked, transferred, or split	Should
FR-15	Canned responses/templates for common replies	Should
FR-16	Resolved tickets auto-close after a configurable period of client inactivity	Should

7.3 Package and SLA Engine

ID	Requirement	Priority
FR-17	Admin can define service packages (e.g. Basic, Premium, Enterprise) and assign one to each client	Must
FR-18	Each package defines first-response and resolution targets per priority level	Must
FR-19	SLA timers start at ticket creation and respect business hours, holidays, and time zones	Must
FR-20	Timers pause while a ticket is Waiting on Client	Must
FR-21	Warning alert at a configurable threshold (e.g. 75% of SLA elapsed)	Must
FR-22	Automatic escalation to team lead, then manager, on breach or imminent breach	Must
FR-23	Priority-based rules (e.g. Critical tickets from Enterprise clients are flagged immediately)	Should
FR-24	Optional package attributes such as support channels, number of users, and ticket quotas	Could

7.4 Notifications

ID	Requirement	Priority
FR-25	Email notification to clients on creation, status change, reply, and resolution	Must
FR-26	Email and in-app notification to staff on assignment, client reply, and SLA warning or breach	Must
FR-27	Notification templates editable by admin	Should
FR-28	SMS/WhatsApp notification channels	Could (phase 2)

7.5 Dashboard and Reporting

ID	Requirement	Priority
FR-29	Live dashboard: open tickets by status, priority, category, and assignee	Must
FR-30	SLA view: tickets at risk and breached, shown prominently	Must
FR-31	Reports: average first-response and resolution times by client, agent, and package	Must
FR-32	Trend reports: ticket volume over time and recurring issue categories	Should
FR-33	Client satisfaction reports	Should
FR-34	Export reports to CSV/PDF	Should

7.6 Administration and Configuration

ID	Requirement	Priority
FR-35	Manage users, roles, and permissions	Must
FR-36	Manage clients/organisations and their contacts	Must

ID	Requirement	Priority
FR-37	Configure categories, priorities, statuses, custom fields, and workflows	Must
FR-38	Configure branding: logo, colours, product name, email sender, language	Must
FR-39	Enable or disable optional modules per installation (feature flags)	Must
FR-40	Configure business hours, holidays, and time zones	Must
FR-41	Knowledge base / FAQ with suggested articles when a client raises a ticket	Could

7.7 Extensibility and Product Packaging

ID	Requirement	Priority
FR-42	Core product contains no customer-specific logic; customisations are separate modules	Must
FR-43	Documented extension points: custom fields, workflow hooks, integrations, reports	Must
FR-44	Public REST API and webhooks for integration with other systems	Should
FR-45	Licence key validation per installation (validity period, user limits, enabled modules)	Must
FR-46	Containerised packaging for one-command installation	Must
FR-47	Versioned releases with automated, safe database migrations	Must
FR-48	Backup/restore and data export tooling for each installation	Should

8. Non-Functional Requirements

Area	Requirement
Security	Role-based access control; password hashing; MFA support; encryption in transit (HTTPS) and for sensitive data at rest; protection against OWASP Top 10; file-upload scanning
Data isolation	A client user must never see another client's data
Performance	Pages load in under 3 seconds; supports at least 50 concurrent staff users and 500 active clients per installation (to be validated)
Availability	Target 99.5% uptime for hosted installations; graceful handling of email or notification failures
Reliability	SLA timers and escalations run through a background job queue and survive restarts
Usability	Responsive on desktop, tablet, and mobile; a new client can raise a ticket in under 2 minutes without training
Auditability	Immutable audit log of ticket and configuration changes
Maintainability	Modular codebase, automated tests, API documentation, and install/upgrade guides
Localisation	Multi-language and time-zone support; configurable date formats
Compliance	Data-protection practices aligned with applicable privacy law (e.g. India's DPDP Act); data retention settings

9. Configuration versus Customisation Model

To keep 80-90% of the product identical across customers:

- **Configuration (no code, admin screen or config file):** packages, SLA targets, categories, priorities, statuses, roles, branding, templates, business hours, language.

- **Modules (separate code plugged into the core):** special integrations, extra data fields with logic, custom reports, customer-specific workflows.
- **Core (never edited per customer):** authentication, ticket engine, SLA engine, notifications, dashboard, admin, API.

Any request that would require editing the core is evaluated and, if broadly useful, added to the core for all customers.

10. Deployment, Licensing and Commercial Model

- **Deployment options:** (a) customer's own server using the container package, or (b) a dedicated instance hosted and managed by us.
- **Licensing:** per-installation licence key; contract grants the right to use, not to resell or redistribute; builds shipped as containers/compiled artefacts.
- **Revenue model (proposed):** one-time or annual licence fee, setup/customisation fee, and annual maintenance and support contract covering updates.
- **Legal:** licence agreement and terms to be drafted by a qualified lawyer.

11. Assumptions and Constraints

- Clients have email access and a web browser; WhatsApp requires a business API provider and approval.
- The initial installation is for our own business and serves as the reference implementation.
- Development uses a widely available web stack so that hiring is straightforward.
- Each installation has its own database; no data is shared between customers.
- Budget, team size, and timeline are to be confirmed.

12. Risks and Mitigations

Risk	Impact	Mitigation
Customisation creeps into the core and the codebase forks per customer	High	Strict module architecture; configuration-first policy; code reviews

Risk	Impact	Mitigation
Updating many separate installations is slow and error-prone	High	Versioned releases, automated migrations, supported-version policy
Strong competition from Freshdesk, Zendesk and Zoho Desk	High	Focus on a niche, data ownership, package-based SLAs, tailoring
Software copied or shared without payment	Medium	Licence keys, containerised builds, legal agreement
SLA calculation errors (business hours, pauses, time zones)	High	Dedicated test suite for the SLA engine; pilot on our own tickets
Staff do not adopt the system and continue using phone calls	High	Make the system the only recognised channel; log phone calls as tickets; management follow-through

13. Release Plan

Phase	Content	Outcome
Phase 1: Internal MVP	Client portal, email ticketing, ticket lifecycle, assignment, package SLAs with alerts, basic dashboard, admin configuration	Live use in our own business
Phase 2: Product hardening	Reports, WhatsApp channel, module system, API, licence keys, Docker packaging, migrations, branding	Installable by an external customer
Phase 3: Pilot and launch	Pilot with 3-5 similar businesses, documentation, support process, pricing, then public launch	First paying customers

14. Success Metrics (KPIs)

- Percentage of issues logged as tickets (target 100%)
- SLA compliance rate by package (target at least 95%)
- Average first-response and resolution time

- Client satisfaction score
- Time to install and configure for a new customer (target under 2 days)
- Share of customer requirements met through configuration alone (target at least 80%)

15. Open Questions

1. Which type of business is the first target market for sale (IT services, clinics, agencies, property management, other)?
2. Will customers run the software on their own servers, or will we host a dedicated instance for each?
3. Which channels are essential for launch: email only, or WhatsApp as well?
4. Do SLAs need to account for business hours and holidays from the first release?
5. What are the budget, team size, and target go-live date?
6. Which existing systems (CRM, accounting, email provider) need integration?
7. What are the data residency and compliance requirements of the target customers?

16. Approval

Name	Role	Signature	Date
	Business Owner		
	Project Manager		
	Technical Lead		